CE80362: Circuit Analysis

Lecture 3 (Continuation): Tellegen's Theorem

Date: 10/12/2026

3.1 Objectives

By the end of this section, you should be able to:

- State Tellegen's theorem correctly.
- Explain the difference between Tellegen's theorem and other network theorems.
- Apply Tellegen's theorem to verify KVL/KCL results.
- Solve and verify examples using Tellegen's theorem.
- Discuss the applications of Tellegen's theorem in circuit network analysis.

3.2 Introduction

Tellegen's theorem was first published in 1952 by Dutch electrical engineer Bernard D.H. Tellegen. It is one of the most general theorems in electrical network theory. Unlike Superposition, Thévenin's, Norton's, or the Maximum Power Transfer Theorem, Tellegen's theorem is not restricted to linear, time-invariant, or passive networks. It depends only on the network topology (the way elements are connected) and Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL).

3.3 Statement of Tellegen's Theorem

"For any electrical network consisting of lumped elements, at any instant of time, the algebraic sum of the products of the branch voltages and branch currents is zero, provided the reference directions of voltages and currents are assigned arbitrarily but consistently with a chosen reference."

Mathematically:

$$\sum_{k=1}^{b} v_k \, i_k = 0$$

Where:

- b = number of branches
- v_k = voltage across branch k with its chosen reference polarity
- i_k = current through branch k with reference direction consistent with the assigned polarity

3.4 Conditions for Application

- The network must be a lumped network obeying KCL and KVL.
- Reference directions for branch currents and polarities for branch voltages may be chosen arbitrarily.
- The theorem holds for linear or non-linear elements.
- It holds for time-invariant or time-varying elements.
- It holds for active and passive elements.
- It is independent of the element constitutive relationships.

3.5 Distinction from Power Conservation

If passive sign convention (current entering positive terminal) is used for all elements, then $v_k i_k$ is the power absorbed by element k. Tellegen's theorem then gives $\sum p_k = 0$, meaning the sum of power absorbed by all passive elements equals the power supplied by all active elements. However, Tellegen's theorem is more general - reference directions need not follow passive sign convention as long as they are consistent.

3.6 Solved Examples

Example 1: Simple Series Circuit

Consider a voltage source V_s in series with R_1 and R_2, supplying current I (loop current).

Branch 1 (Source): $v_1 = V_s$, $i_1 = I$ (in a consistent reference direction)

Branch 2 (R_1): $v_2 = I R_1$ (voltage drop in direction of I)

Branch 3 (R_2): $v_3 = I R_2$ (voltage drop in direction of I)

Applying Tellegen's theorem:

$\sum v_k i_k = v_1 i_1 + v_2 i_2 + v_3 i_3 = V_s I - I^2 R_1 - I^2 R_2 = I(V_s - I(R_1+R_2)) = I(0) = 0$

Example 2: Two-Loop Network (Verification)

For any two-loop (or general) network where branch voltages and currents have been found satisfying KVL and KCL with consistent references, taking the algebraic sum $\sum v_k i_k$ over all branches gives zero. This provides a quick and reliable check of calculations.

3.7 Comparison with Other Network Theorems

Theorem	Basis	Applies To	Purpose	Main Limitation/ Note

Superposition	Linearity	Linear, bilateral	Find I/V due to individual sources	Does not apply directly to power
Thévenin's	Linearity & equivalence	Linear, bilateral (two-terminal)	Reduce network to V_{th} in series with R_{th}	Valid for external load; not a general verification tool
Norton's	Current source equivalence	Linear, bilateral (two-terminal)	Reduce to I_N in parallel with R_N	Dual of Thévenin's theorem
Maximum Power Transfer	Load matching	Linear	Find R_L for maximum power transfer	Efficiency is 50% at maximum power
Tellegen's	KCL/KVL & topology	All lumped networks (linear/ nonlinear, time-varying)	Verification, relations between networks, theoretical foundation	Does not solve for unknowns alone; needs KVL/KCL

3.8 Applications in Circuit Network Analysis

- **Verification:** Check that computed branch voltages and currents satisfy the network equations.
- **Power balance:** Verify total power supplied by sources equals total power absorbed by elements.
- **Reciprocity theorems:** Used to derive and prove reciprocity in linear time-invariant networks.
- **Adjoint networks & sensitivity analysis:** Essential in network theory, optimization and circuit design.
- **Network equivalence & topology studies:** Relates different networks that share the same topology/graph.
- **Analysis of nonlinear/time-varying circuits:** Applicable where theorems like Superposition and Thévenin's do not hold.

3.9 Assignment (End of Lecture 3)

Question: By giving examples, do a detailed research and discuss the tellegen's theorem used in circuit network analysis

Solution

Definition: Tellegen's theorem states that for any lumped electrical network, $\sum_{k=1}^{b} v_k i_k = 0$ at every instant, provided branch voltages and currents are assigned with consistent references and the network satisfies KCL and KVL.

Examples: In a simple series circuit, the source power equals the sum of power dissipated in resistors. In a multi-branch network, summing $v_k i_k$ over all branches gives zero regardless of whether elements are linear or nonlinear.

Discussion: Tellegen's theorem is topology-based and extremely general. Unlike other network theorems used for reduction or source removal, it is primarily a verification and theoretical theorem. Its applications extend to proving reciprocity, sensitivity analysis, and working with nonlinear/time-varying networks.

3.10 Conclusion

Tellegen's theorem is a fundamental conservation law of electrical network theory. While it does not directly solve for unknown voltages or currents like Nodal/Mesh analysis, or reduce networks like Thévenin's/Norton's theorems, it is indispensable for verification and forms the theoretical basis for many important network theorems. Its generality makes it one of the most powerful and far-reaching results in circuit analysis.

References

- Tellegen, B.D.H. (1952). "A general network theorem, with applications". Philips Research Reports.
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- Boylestad, R.L., *Introductory Circuit Analysis*.
- Alexander, C.K. & Sadiku, M.N.O., *Fundamentals of Electric Circuits*.